Database Design Document

Project

Community Library Management System

Database Technology

- PostgreSQL
- Entity Framework Core (Npgsql Provider)

Tables

membershiptypes

Purpose:
Stores membership plans and borrowing limits.

Attributes:

- id (Primary Key)
- name (Unique, Not Null)
- maximumborrowings (Not Null)
- maximumborrowdays (Not Null)

Seed Data:

- Basic → 2 books → 7 days
- Student → 3 books → 10 days
- Premium → 5 books → 15 days

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members

Purpose:
Stores member information and account status.

Attributes:

- id (Primary Key)
- name (Not Null)
- phone (Unique, Indexed, Not Null)
- email (Unique, Indexed, Not Null)
- accountstatus (Default: Active, Not Null)
- membershiptypeid (Foreign Key → membershiptypes.id, OnDelete Restrict)

Possible Account Status Values:

- Active
- Inactive
- Deactivated

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bookcategories

Purpose:
Stores book category information.

Attributes:

- id (Primary Key)
- name (Unique, Not Null)

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books

Purpose:
Stores book details using ISBN.

Attributes:

- isbn (Primary Key)
- title (Indexed, Not Null)
- authorname (Indexed, Not Null)
- edition
- price (Not Null)
- categoryid (Foreign Key → bookcategories.id, OnDelete Restrict)

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bookcopies

Purpose:
Stores physical copies of books.

Attributes:

- barcodeno (Primary Key)
- isbn (Foreign Key → books.isbn)
- status (Indexed, Default: Available, Not Null)

Possible Status Values:

- Available
- Borrowed
- Damaged
- Unavailable

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borrowings

Purpose:
Tracks borrowing and return transactions.

Attributes:

- id (Primary Key)
- memberid (Foreign Key → members.id, Indexed, OnDelete Restrict)
- barcodeno (Foreign Key → bookcopies.barcodeno, OnDelete Restrict)
- borrowdate (Not Null)
- duedate (Not Null)
- returndate
- borrowstatus (Default: Borrowed, Not Null)

Possible Borrow Status Values:

- Borrowed
- Returned

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fines

Purpose:
Stores fines generated during borrowing operations.

Attributes:

- id (Primary Key)
- borrowingid (Foreign Key → borrowings.id, Indexed)
- finetype (Not Null)
- fineamount (Not Null)
- ispaid (Default: false, Not Null)

Possible Fine Types:

- LateReturn
- Damage
- Other

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finepayments

Purpose:
Tracks fine payment transactions.

Attributes:

- id (Primary Key)
- fineid (Foreign Key → fines.id)
- amountpaid (Not Null)
- paymentdate (Default: CURRENT_TIMESTAMP, Not Null)

Relationships

- One membershiptype can have many members
- One bookcategory can have many books
- One book can have many bookcopies
- One member can have many borrowings
- One bookcopy can have many borrowings
- One borrowing can have many fines
- One fine can have many finepayments

Indexes

Indexes are created on:

- books.title
- books.authorname
- members.phone
- members.email
- bookcopies.status
- borrowings.memberid
- fines.borrowingid

Business Rules

- Only active members can borrow books
- Borrowing limits depend on membership type
- Members with unpaid fines above ₹500 cannot borrow books
- Only available book copies can be borrowed
- Damaged or unavailable copies cannot be borrowed
- Fine for late return is ₹10 per delayed day

Stored Functions and Procedures

- create_borrowing(memberId, barcodeno)
- return_book(barcodeno)
- calculate_member_fine(memberId)
- get_available_books_by_category(categoryId)
- get_member_borrowing_summary(memberId)

Technical Notes

- Entity Framework Core is used for ORM operations
- PostgreSQL stores enum values as strings
- Transactions are used during borrowing and return workflows

- Foreign key restrictions are used to protect historical records

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Conclusion

The database design provides a structured and normalized approach for handling library operations such as member management, borrowing, returns, fines, and reporting. The schema is designed to maintain data integrity while supporting future scalability and feature enhancements.